

Outline

1. Fujikawa's Path Integral Technique
 - (i) The Measure
 - (ii) Computing the anomaly
 - (iii) $\pi^0 \rightarrow \gamma\gamma$
2. General Chiral Anomalies
 - (i) Structure of anomaly diagrams
 - (ii) Categorization

1. Fujikawa's Path Integral Technique

Work in Euclidean space

$$\mathcal{L} = \bar{\psi} (i\not{D} - m) \psi - \frac{1}{2} \text{Tr}_G F_{\mu\nu} F^{\mu\nu}$$

$\uparrow$
group indices

$$\not{D} = \gamma^\mu (\partial_\mu - ig A_\mu^a T^a)$$

$\underbrace{\hspace{1.5cm}}_{\text{generate gauge group}}$

(i) How do we define the path integral measure?

Must Choose a basis for $\psi, \bar{\psi}$

In Euclidean space, $i\not{D}$ is hermitian

define basis ϕ_k s.t. $i\not{D}\phi_k = \lambda_k \phi_k$

$$\int d^4x \bar{\phi}_n(x) \phi_m(x) = \delta_{nm}$$

$$\sum_k \phi_k(x) \bar{\phi}_k(y) = \delta^4(x-y)$$

$$\psi(x) = \sum_k a_k \phi_k(x) \quad \Bigg| \quad \bar{\psi}(x) = \sum_k \bar{b}_k \bar{\phi}_k(x)$$

$a_k \& \bar{b}_k$ grassmann numbers

The path integral measure becomes

$$\int \prod \bar{\psi} \psi \rightarrow \prod_n \int d\bar{b}_n da_n$$

We want to study transformations of the form

$$\begin{aligned}\psi(x) &\rightarrow u(x) \psi(x) \\ \bar{\psi}(x) = \psi^\dagger \gamma^4 &\rightarrow \underbrace{\psi^\dagger u^\dagger}_{\gamma^4 \cdot \gamma^4} \gamma^4 \\ &= \bar{\psi}(x) \gamma^4 u^\dagger(x) \gamma^4\end{aligned}$$

How about $a_n \& b_n$

$$\psi(x) = \sum_n a_n \phi_n(x) \rightarrow \sum_n a_n u(x) \phi_n(x)$$

by orthonormality,

$$\int d^4x \bar{\phi}_m(x) \psi(x) = a_m(x)$$

$$a'_m = \left[\underbrace{\int d^4x \bar{\phi}_m(x) u(x) \phi_n(x)}_{\bar{u}_{mn}} \right] a_n$$

likewise

$$b'_m = \bar{u}_{mn} b_n$$

$$\bar{u}_{mn} = \int d^4x \bar{\phi}_n(x) \gamma^4 u^\dagger \gamma^4 \phi_m(x)$$

Ordinarily, measure transforms by $\det\{\text{Jacobian}\}$. However grassmann integration is more akin to differentiation. It also anticommutes

$$\prod_n \int d\bar{b}_n da_n \sim \prod_n \frac{\partial}{\partial b_n} \frac{\partial}{\partial a_n}$$

so

$$\rightarrow \prod_n d\bar{b}'_n da'_n (\det u \det \bar{u})^{-1}$$

First consider classical symmetry transformations treating LH & RH particles the same

$$u = \exp\{i\alpha(x)t\} \quad [t, \gamma^\mu] = 0$$

$\uparrow$
symmetry generator

$$\simeq 1 + i\alpha(x)t$$

$$\det u = \det \left(1 + i \int d^4x \alpha(x) \bar{\psi}_m(x) t \psi_m(x) \right)$$

$$= \det \left(e^{i \int d^4x \alpha(x) \bar{\psi}_m(x) t \psi_m(x)} \right)$$

$$= \exp \left(i \sum_{\kappa} \left(\int d^4x \alpha(x) \bar{\psi}_{\kappa}(x) t \psi_{\kappa}(x) \right) \right)$$

$\uparrow$
Trace over $m \& n$

$$\det \bar{u} = \exp \left(-i \sum_{\kappa} \int d^4x \alpha(x) \bar{\psi}_{\kappa}(x) \overleftarrow{\gamma^4} t \overrightarrow{\gamma^4} \psi_{\kappa}(x) \right)$$

$(\gamma^4)^2 = 1$

$$= \exp \left(-i \sum_{\kappa} \int d^4x \alpha(x) \bar{\psi}_{\kappa}(x) t \psi_{\kappa}(x) \right)$$

$$\det u \det \bar{u} = 1$$

The path integral measure is invariant! No anomaly!

Now consider chiral symmetries, which treat LH & RH particles differently. Consider the subset

$$u(x) = \exp(i\alpha(x)\gamma_5 t) = 1 + i\alpha(x)\gamma_5 t$$

LH & RH particles rotate oppositely

$$\det u = e^{i \sum_{\kappa} \left(\int d^4x \alpha(x) \bar{\psi}_{\kappa}(x) \gamma_5 t \psi_{\kappa}(x) \right)}$$

$$\det \bar{u} = \exp \left(+i \sum_k \int d^4x \alpha(x) \bar{\phi}_k(x) \overbrace{\gamma^4 \gamma_5}^{= +\gamma_5 \gamma^4} \phi_k(x) \right)$$

$$= \det u$$

$$[\det u \det \bar{u}]^{-1} = (\det u)^{-2}$$

$$= \exp \left(- \int d^4x \alpha(x) \mathcal{A}(x) \right)$$

$$\boxed{\mathcal{A}(x) = 2i \sum_k \bar{\phi}_k(x) \gamma_5 \phi_k(x)}$$

This is the anomaly function. We will use it to compute $2\pi \mathcal{J}_5^\mu$ & $\Gamma(\pi^0 \rightarrow 2\gamma)$.

(ii) Let's compute the anomaly function

Using the completeness relation,

$$\mathcal{A}(x) = 2i \text{Tr}_{\text{Dirac indices}} \left\{ +\gamma_5 \sum_k \phi_k(x) \bar{\phi}_k(x) \right\}$$

$$= 2i \underbrace{\text{Tr}_{\text{Dirac indices}} \{ +\gamma_5 \}}_{\text{but } \text{Tr}_D \gamma_5 = 0} \underbrace{\delta^4(x-x)}_{\text{infinite}}$$

ill-defined!
Must regulate!

Naive choice: Use a momentum regulator

$$\text{write } \delta^4(x-y) = \int \frac{d^4p}{(2\pi)^4} \left\{ u_s(p) \bar{u}_s(p) e^{ip \cdot (x-y)} + v_s(p) \bar{v}_s(p) e^{ip \cdot (y-x)} \right\}$$

$$\bar{u}_s(p) \gamma_5 u_s(p) = \bar{v}_s(p) \gamma_5 v_s(p) = 0$$

$A(x)$ vanishes!

but, $\int \frac{d^4 p}{(2\pi)^4}$ is ∞

This is akin to saying $\sum_{n=0}^{\infty} (-1)^n = 0$. No!
We must regulate this sum to claim it vanishes.

We can use a momentum cutoff Λ or a regulator like $e^{\not{p}^2/\Lambda^2} = e^{-p^2/\Lambda^2}$

Problem: Momentum regulators do not respect gauge invariance! We can only set the axial anomaly to 0 in this way by violating gauge invariance.

Gauge Invariant Regulator:

$$\begin{aligned} A(x) &= 2i \lim_{\substack{\Lambda \rightarrow \infty \\ x \rightarrow y}} \text{Tr}_{\text{GO}} [\gamma_5 e^{\not{p}^2/\Lambda^2}] \delta^4(x-y) \\ &= 2i \lim_{\Lambda \rightarrow \infty} \sum_k \bar{\phi}_k(x) \gamma_5 e^{-k^2/\Lambda^2} \phi_k(x) \end{aligned}$$

Manifestly convergent!

How do we evaluate this?

(Schematically)

$$1. \quad A(x) = 2i \lim_{\Lambda \rightarrow \infty} \int \frac{d^4 k}{(2\pi)^4} \text{Tr}_{\text{GO}} [\gamma_5 e^{\not{p}^2/\Lambda^2}] e^{ik \cdot (x-y)} \Big|_{x=y}$$

pulls out k

set $x=y$
to get rid
of exp.

2. The only nonzero Tr_D with a γ_5 in is

$$\text{Tr}_D[\gamma_5 \gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma] = 4\epsilon^{\mu\nu\rho\sigma}$$

this selects out the piece

$$A(x) = i \int \frac{d^4k}{(2\pi)^4} e^{-k^2} \text{Tr}_D[\gamma_5 \not{x}]$$

$$3. \not{D}^2 = D^2 + \frac{1}{2} \gamma^\mu \gamma^\nu \underbrace{\{D_\mu, D_\nu\}}_{F_{\mu\nu}!}$$

$$A(x) = \frac{-ig^2}{16\pi^2} \epsilon^{\mu\nu\rho\sigma} \text{Tr}_G[F_{\mu\nu} F_{\rho\sigma}]$$

write $F_{\mu\nu} = F_{\mu\nu}^a t^a \leftarrow$ generators of gauge group

$$A(x) = \frac{-ig^2}{16\pi^2} \epsilon^{\mu\nu\rho\sigma} F_{\mu\nu}^a F_{\rho\sigma}^b \text{Tr}_G[t^a t^b]$$

(iii) Now, let's compute $2\pi J_5^M \rightarrow \Gamma(\pi^0 \rightarrow 2\gamma)$

Murali showed that under symmetry transformation U ,

$$\int D\bar{\psi} D\psi e^{-S[\bar{\psi}, \psi]}$$

$$\text{Murali} \rightarrow \int D\bar{\psi} D\psi e^{-S[\bar{\psi}', \psi']} + \int d^4x \alpha(x) 2\pi J_5^M$$

$$\text{Accounting for measure} \rightarrow \int D\bar{\psi} D\psi' e^{-S[\bar{\psi}', \psi']} + \int d^4x \alpha(x) [2\pi J_5^M - A(x)]$$

i goes away when going from
Euclidean to Minkowski space

$$\therefore \boxed{\mathcal{L}_5(x) = - \frac{g^2}{16\pi^2} \epsilon^{\mu\nu\rho\sigma} F_{\mu\nu}^a F_{\rho\sigma}^b \text{Tr}\{t_a t_b t_c\}}$$

$\epsilon^{\mu\nu\rho\sigma}$ does not transform under Wick rotation.

Consider $\pi^0 \rightarrow 2\gamma$

π^0 is the goldstone boson associated to the spontaneously broken $SU(2)_L \times SU(2)_R \rightarrow SU(2)_V$ (approximate) symmetry of the strong interactions.

π^0 is associated to the broken generator $t = \gamma^5 \tau^3$

$$\tau_3 = \begin{pmatrix} 1 & \\ & -1 \end{pmatrix}$$

$$\begin{aligned} u &\rightarrow i\alpha \gamma_5 u \\ d &\rightarrow -i\alpha \gamma_5 d \end{aligned}$$

There is no anomaly in QCD since u & d transform under the same irreducible representations of $SU(3)_c \rightarrow t_a = t_b \propto \mathbb{1}$ (in flavor space)

However, In QED, $t_a = t_b = q = \begin{pmatrix} 2/3 & \\ & -1/3 \end{pmatrix}$

$$\begin{aligned} \therefore \text{Tr}_G \{ q^2 \tau_3 \} &= N_c \left\{ \left(\frac{2}{3}\right)^2 \cdot (+1) + \left(-\frac{1}{3}\right)^2 \cdot (-1) \right\} \\ &= \frac{N_c}{3} \end{aligned}$$

$$\therefore \mathcal{L}(x) = - \frac{g^2 N_c}{48\pi^2} \epsilon^{\mu\nu\rho\sigma} F_{\mu\nu} F_{\rho\sigma}$$

How do we couple to the π^0 ? From the EFT,

$$\pi^0 \rightarrow \pi^0 + \delta\pi^0 \quad \left| \quad \delta\pi^0 = \alpha(x) \underbrace{F_\pi}_{\text{pion decay constant}} \right.$$

Want $\mathcal{L}_{eff} \rightarrow \mathcal{L}_{eff} + \delta \mathcal{L}_{eff} \mid \delta \mathcal{L}_{eff} = \alpha(x) A(x)$

choose $\mathcal{L}_{eff} \supset \frac{\pi^0 A(x)}{F_\pi}$

$$\therefore \delta \mathcal{L}_{eff} = \frac{\delta \pi^0 A(x)}{F_\pi} = \frac{\alpha(x) A(x) \cancel{F_\pi}}{\cancel{F_\pi}}$$

$$\therefore \boxed{\mathcal{L}_{eff} \supset - \frac{g^2 N_c}{48 \pi^2 F_\pi} \epsilon^{\mu\nu\rho\sigma} F_{\mu\nu} F_{\rho\sigma} \pi^0}$$

$\parallel$
 g_{eff}

$$\Gamma(\pi^0 \rightarrow 2\gamma) = \frac{m_\pi^3 g_{eff}^2}{\pi} \stackrel{N_c=3}{=} 1.1 \times 10^{16} \text{ s}^{-1}$$

$$\Gamma_{exp}(\pi^0 \rightarrow 2\gamma) = (1.19 \pm 0.08) \times 10^{16} \text{ s}^{-1}$$



2. Direct Computation of Chiral Anomalies

- Package all LH fermions & anti-fermions into a column

- if ψ contains all LH & RH fermions,

$$\chi \equiv \left\{ \begin{array}{c} \frac{1}{2}(1+\gamma_5)\psi \\ \frac{1}{2}(1+\gamma_5)\psi^* \end{array} \right\} \quad \text{charge conjugation}$$

- Study general chiral symmetry transformations of the form

$$\delta\chi = i\epsilon_\alpha T_\alpha \chi$$

↑ can mix LH & RH particles in any way

- e.g. if

$$\delta\psi = i\theta_\alpha \left[\frac{1}{2}(1+\gamma_5)\psi_\alpha^L + \frac{1}{2}(1-\gamma_5)\psi_\alpha^R \right] \psi$$

$$T_\alpha = \left\{ \begin{array}{c} \psi_\alpha^L \\ -\psi_\alpha^{R*} \end{array} \right\} = \left\{ \begin{array}{c} \psi_\alpha^L \\ -(\psi_\alpha^R)^T \end{array} \right\}$$

We will compute the three-point function

$$\Gamma_{abc}^{\mu\nu\rho} \equiv \langle T \{ j_a^\mu(x), j_b^\nu(y), j_c^\rho(z) \} \rangle_{\text{vac}}$$

(i) Structure of Δ 's!

$$i\Gamma_{abc}^{\mu\nu\rho} =$$

The first diagram shows a triangle loop with vertices labeled $j_a^\mu(x)$, $j_b^\nu(y)$, and $j_c^\rho(z)$. The second diagram shows a triangle loop with vertices labeled $j_a^\mu(x)$, $j_b^\nu(y)$, and $j_c^\rho(z)$.

$$i\Gamma_{abc}^{\mu\nu\rho} = \text{Tr}_G [S(x-y) T_b \gamma^\nu P_L S(y-z) T_c \gamma^\rho P_L S(z-x) T_a \gamma^\mu P_L] \\ + \{y \leftrightarrow z, b \leftrightarrow c, \nu \leftrightarrow \rho\}$$

Important Notes on the Structure

1. $i\Gamma_{abc}^{\mu\nu\rho} \propto \text{Tr}_G \{T_b T_c T_a\}$ since T 's don't carry Dirac indices & nothing else carries group indices

$$\rightarrow = D_{abc} + \frac{1}{2} i k_R f_{abc}$$

$\uparrow$ completely symmetric $\uparrow$ completely antisymmetric

$$D_{abc} \equiv \frac{1}{2} \text{Tr}_G \{ \{T_a, T_b\} T_c \}$$

$$\text{Tr}_G \{ [T_a, T_b] T_c \} = i k_R f_{abc}$$

Claim: completely antisymmetric piece does not contribute to the anomaly. They just contribute the contact terms we saw before which do not break any symmetries.

$$\left[\frac{\partial}{\partial x^\mu} \Gamma_{abc}^{\mu\nu\rho} \right]_{\text{formal}} = -i \langle a b d \rangle \delta^4(x-y) \langle j_d^\nu(y) j_c^\rho(z) \rangle_{\text{vac}} \\ - i \langle a c d \rangle \delta^4(x-z) \langle j_b^\nu(y) j_d^\rho(z) \rangle_{\text{vac}}$$

$\therefore$ anomaly of D_{abc}

2. How do we regulate the integral?

- Cannot use dimensional regularization since γ^5 has no counterpart in fractional dimension.
- Cannot use Pauli-Villars since a fermion mass violates all chiral symmetries

• Must use momentum shift V in showed us.

Result:

$$\left[\frac{\partial}{\partial x^\mu} \Gamma_{abc}^{\text{vnp}}(x, y, z) \right] = \frac{2}{(2\pi)^{12}} \text{Dabc} \int d^4k_1 d^4k_2 e^{-i(k_1+k_2)x} e^{ik_1 \cdot y} e^{ik_2 \cdot z} \pi^2 \epsilon^{\mu\nu\lambda\rho} p_{a\mu}(k_1+k_2)_\lambda$$

Let's talk about Dabc!

Dabc is the only piece of this calculation that depends on the particular symmetries we are studying. It tells us everything!

e.g.

If $T_a = \begin{pmatrix} t_a^L \\ -(t_a^R)^T \end{pmatrix}$

$$\begin{aligned} \text{Dabc} &= \frac{1}{2} \text{Tr}_G \left[\{t_a^L, t_b^L\} t_c^L \right] - \frac{1}{2} \text{Tr}_G \left[\{t_a^R, t_b^R\} t_c^R \right] \\ &= \text{Dabc}(L) - \text{Dabc}(R) \end{aligned}$$

Consider when all t_a 's are vector (V) or axial vector (A) transformations

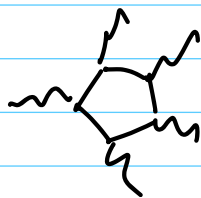
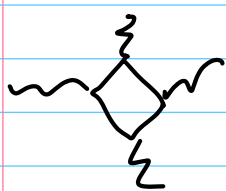
$$\begin{aligned} V &\rightarrow t_a^L = t_a^R \\ A &\rightarrow t_a^L = -t_a^R \end{aligned}$$

Clear that VVV & VAA anomaly diagrams vanish. VVA & AAA anomaly diagrams are nonvanishing, in general, &

$$\text{Dabc} = \text{Tr}_G \left[\{t_a^L, t_b^L\} t_c^L \right]$$

What else contributes to anomalies?

 $\rightarrow$ evaluate in dimreg $\rightarrow$ no problem!

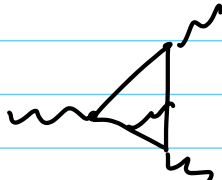


$\rightarrow$ These diagrams do contribute! (More on them later!)

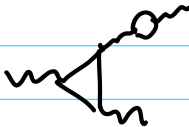
Higher N -gons diagrams are convergent and do not contribute to the anomaly.

What about more loops?

- Adler & Bardeen derived anomaly to all orders in PT. They found that chiral anomalies are one loop exact!

gist: diagrams like  don't contribute

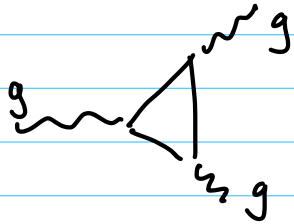
because they reduce the mass dimensionality of the integral, so they do not contribute

- diagrams like  do contribute, but they just renormalize operators in our original anomaly diagram

(ii) Categorization:

The physical significance of an anomaly diagram depends on what kinds of symmetries, **gauge** (local) or **flavor** (global), it couples to.

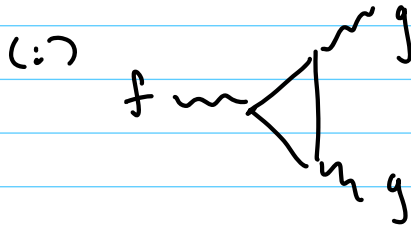
1. gauge anomalies



These diagrams must all vanish or the gauge theory is sick!
 (Gauge symmetries are not true symmetries; they are definitional redundancies.)

Matthew will discuss these

2. mixed anomalies



This looks like the axial anomaly that we've been studying!

Pick $a_\mu = (k_1 - k_2)_\mu$ so gauge symmetries are conserved.

$$\left\{ \frac{\partial}{\partial x^\mu} \Gamma_{abc}^{\mu\nu\rho} \right\}_{\text{anom}} = -\frac{1}{4\pi^2} D_{abc} \epsilon^{\kappa\nu\lambda\rho} \frac{\partial \delta^4(y-x)}{\partial y^\kappa} \frac{\partial \delta^4(z-x)}{\partial z^\lambda}$$

We can couple gauge symmetry currents to gauge fields!

$$\langle J_a^\mu(x) \rangle = \frac{1}{2} \int d^4y d^4z \Gamma_{abc}^{\mu\nu\rho} A_\nu^b(y) A_\rho^c(z)$$

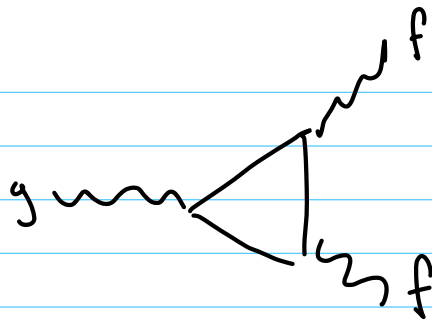
$$\left[\langle \partial_\mu J_a^\mu(x) \rangle \right]_{\text{anom}} = -\frac{1}{8\pi^2} D_{abc} \epsilon^{\kappa\nu\lambda\rho} \partial_\kappa A_\nu^b \partial_\lambda A_\rho^c$$

Not gauge invariant? The square & pentagon diagrams must add up to give the gauge invariant result

$$\boxed{\int \langle \partial_\mu J_a^\mu(x) \rangle_{\text{anom}} = -\frac{1}{32\pi^2} D_{abc} \epsilon^{\kappa\nu\lambda\rho} F_{\kappa\nu}^b F_{\lambda\rho}^c}$$

→ agrees with Fujikawa for $UV A$ anomaly

(ii)



I don't know much about these.

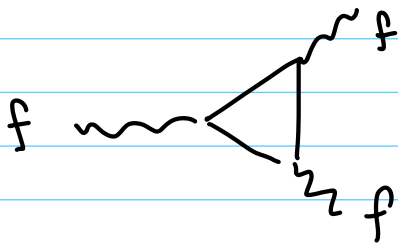
$$D_{abc} = \frac{1}{2} \text{Tr}_G [T_a \{T_b, T_c\}]$$
$$= \underbrace{\text{Tr}_g [T_a]}_{\text{only non zero}} \text{Tr}_f [T_b T_c]$$

only non zero

if gauge group
is $u(1)$ - $su(N)$

generators are
traceless

3. t'Hooft anomalies



Cullen will discuss these
Anomalies for flavor
symmetries are interesting, not
fatal. Can use to understand
IR spectrum of strongly
coupled gauge theories.

